Loads

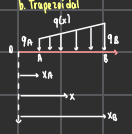
- a. Static Loading: Application time for the load is considerably larger than the period(T).
- b. Dynamic Loading: When the application of the load is time-varying.

* Loads can also be classified in 2 sub-groups: point loads and distributed loads. However, in reality, we don't see point loads. In certain situations, we may assume some loads to be point loads.

Distributed Loads
- Regularly Distributed Loads
  - a. Uniformly
  - b. Trapezoidal
  - c. Triangular
  - d. Parabolic
- Irregularly Distributed Loads

$$\Theta = \sum_{i=1}^{n} q(x_i)\,dx$$

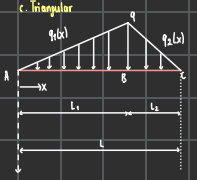
b. Trapezoidal



$$q(x) = \frac{q_B - q_A}{x_B - x_A}\,x + \frac{q_A x_B - q_B x_A}{x_B - x_A}$$

if $x_A = 0$ and $x_B = L$, $q(x) = \frac{q_B - q_A}{L}\,x + q_A$

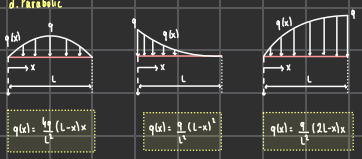
c. Triangular



$$q_1(x) = \frac{q}{L_1}\,x$$

$$q_2(x) = q\,\frac{L - x}{L_2}$$

d. Parabolic



$$q(x) = \frac{4q}{L^2}(L - x)x$$

$$q(x) = \frac{q}{L^2}(L - x)^2$$

$$q(x) = \frac{q}{L^2}(2L - x)x$$

Location of the Resultant of the Distributed Load

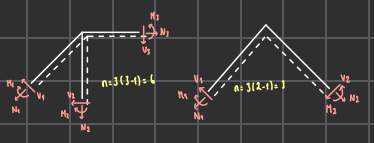
Moments with respect to O:

1. Regularly D.L., $x_{\Theta} \int_{x_A}^{x_B} q(x)\,dx = \int_{x_A}^{x_B} q(x)\,x\,dx \rightarrow x_{\Theta} = \frac{\int_{x_A}^{x_B} q(x)\,x\,dx}{\int_{x_A}^{x_B} q(x)\,dx}$

2. Irregularly D.L., $x_{\Theta} \sum \Delta\Theta_i = \sum (\Delta\Theta_i\, x) \rightarrow x_{\Theta} = \frac{\sum (\Delta\Theta_i\, x)}{\sum \Delta\Theta_i}$

Internal loads at the joints of planar systems

1. Fixed (Rigid) Joint: $n = 3(n_d - 1)$ (unknowns)



2. Pin Joints: Moments don't occur on these joints.

unknowns: $n = 2(n_d - 1)$

3. Axial Hinge Joint: Unknowns are shear force and bending moment.

4. Shear Hinge Joint: Unknowns are axial force and bending moment.

Instability

1. Partial Constraints: Structure or its members have fewer reaction forces than equations of equilibrium.
2. Improper Constraints: Either the line of action of the reactive forces intersect at a common point or are parallel to one another.



Indeterminacy: When the number of unknown forces are greater than number of equilibrium equations, the structure is statically indeterminate.

→ Hinge (Moment Release) Condition: Each hinge reduces the degree of indeterminacy by 1.

Determining the Degree of Indeterminacy

a. Separating structures into two parts with a single cut



n_r: number of the unknown support reactions
n_m: additional equations with hinges
n_{el}: element number connected to hinge

$$n_m = n_{el} - 1 \qquad n = n_r - (n_m + 3)$$

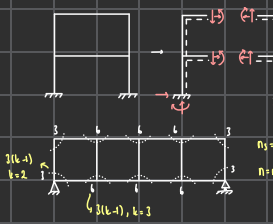
$n_r = 10$, $n_m = 3$, $n = 10 - (3+3) = 4$

$n = 0$	Statically determinate
$n < 0$	unstable
$n > 0$	Statically indeterminate to the n-th degree

b. Method of Sections: Each part should be in equilibrium under external loads at the cut section.

n_s: number of separated parts
n_r: total number of support reactions
n_{ik}: number of unknown internal forces at the cut section

$$n = n_{ik} + n_r - 3n_s$$



$n_s = 2$, $n_r = 6$, $n_{ik} = 6$, $n = 6 + 6 - 3(2) = 6$

$n_s = 13$, $n_r = 3$, $n_{ik} = 48$, $n = n_{ik} + n_r - 3n_s = 12$

→ This structure is externally determinate yet internally indeterminate.
(Support reactions can be found using eq of equil.)

c. General Methods

s: number of structural elements
j: number of joints including the supports
n_r: number of support reactions
n_m: number of hinge condition

total number of equilibrium equations: $n_{eq} = 2j + 3j$
total number of forces: $n_f = b_s + n_r$

$$n = n_f - n_{eq} = 3(s - j) + n_r$$
$$n = 3(1 - j) + n_r - n_m$$

d. Trusses: $n_{eq} = 2j$, $n_f = s + n_r$ $\Rightarrow$ $n = n_f - n_{eq} = (1 - 2j) + n_r$